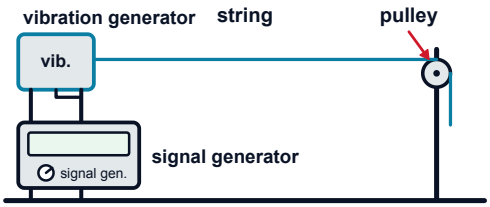
DO NOW · REQUIRED PRACTICAL

Waves on a string

Waves
Paper 2 · Combined and Separate
6 steps

A student measures the speed of a wave on a stretched string.

1 Attach one end of a string to a vibration generator and pass the other end over a pulley.



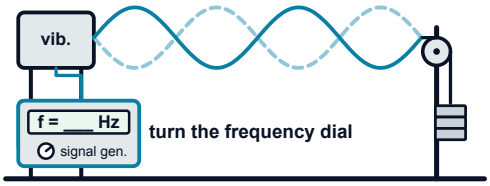
2 Hang masses on the end of the string to keep it taut.



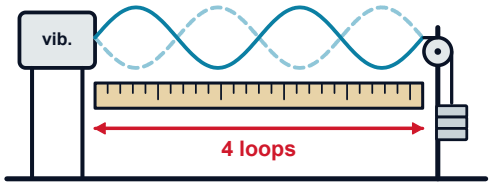
masses keep the string taut

3 Switch on the signal generator and adjust the frequency until clear loops appear.

adjust the frequency until clear loops appear

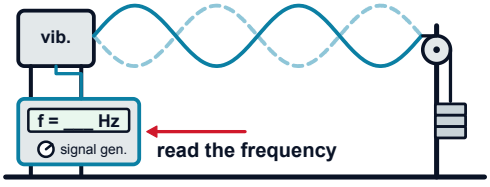


4 Measure the length of several loops and work out the wavelength.



1 loop = ½ λ, so λ = length of 4 loops ÷ 2

5 Read the frequency from the signal generator.



6 Calculate the wave speed: frequency × wavelength.

$$v = f \times \lambda$$

wave speed = frequency × wavelength

m/s = Hz × m

use f from step 5 and λ from step 4

Note: The frequency can be read before or after the wavelength is measured.

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